



Writing an Interim Project Report

Dr. (Eng.) Anthony Muchiri
19.02.2026



Update on Semester schedule

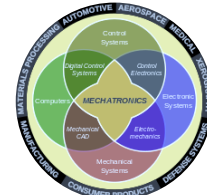
Presentation date: **6th March 2025 (TBC)**

Submit your draft reports to your supervisors by
27th February 2026

Final report (2 copies) should be printed, bound,
signed by students and supervisors by
5th March 2026

Outline – Interim Report

- Format
- Abstract
- Introduction (Chapter 1)
- Literature Review (Chapter 2)
- Methodology (Chapter 3)
- Results & Discussion (Chapter 4)
- Conclusion: (Chapter 5)
- References
- Formating: Equations, Figures, Tables
- Some Do's and Don'ts
- Oral Presentation assessment
- Mark sheet templates

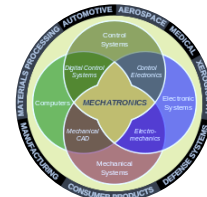


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Report Format

- Title Page
- Declaration
- Abstract
- TOC
- Nomenclature
- Introduction
- Literature Review
- Methodology
- **Results and Discussion**
- Conclusion
- References
- Appendices



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Report Format

Title Page

☐ University Logo (the correct one!!!),
College/School/Department

☐ Title: Brief and suitable for indexing

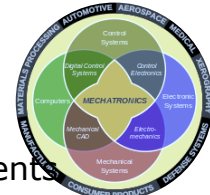
☐ Project Code: FYP-19-xx

☐ Authors

☐ Supervisors

☐ Date

☐ Statement on fulfilment of degree requirements



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Report Format

Title Page



Note the order



Project title

Project Code

Type of report

Jomo Kenyatta University of Agriculture and Technology
College of Engineering and Technology
School of Mechanical, Manufacturing, and Materials Engineering
Department of Mechatronic Engineering

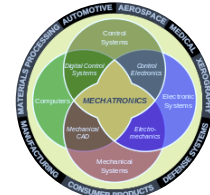
Design and Fabrication of a Delta PCB Milling
Machine

FYP-16-38
INTERIM REPORT

Christopher Mambo (EN293-6060/2015)
Christine Sifa (EN293-6061/2015)

Supervisors
Ms Lucy Karuki
Mr Patrick Kipkosgei

January 2021



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Report Format

Declaration

- ☐ Statement of originality, duly signed by the students, and countersigned by the supervisors.



Abstract

- ☐ A very precise introduction, approach (methodology), results and conclusion
- ☐ No referencing
- ☐ No un-identified abbreviations: e.g. SI units are okay. PWM not okay.
- ☐ No words or abbreviations in brackets

NOTE

- ☐ This is **NOT your project proposal**. DO NOT use the same abstract you used in the proposal.



Table of Contents

- ☐ Chapter, sections, subsections (1, 1.1, 1.1.1)
- ☐ A Table of Figure is optional



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Nomenclature

- ☐ Abbreviations and Symbols, arranged alphabetically.



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Pagination

- ☐ Roman (i, ii,...)before introduction, Arabic (1, 2, ...) thereafter

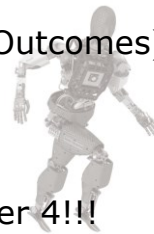


Introduction

- ☐ What is the problem?
- ☐ What is the background information?
- ☐ Introduce the problem and what it aims to achieve
- ☐ State the problem
- ☐ State the objectives (Main, Specific)
- ☐ State what you intend to achieve (Expected Outcomes)
- ☐ **Justification**

NOTE:

Expected outcomes are listed here, not in Chapter 4!!!



Chapter 2: Literature review

☐ How is the problem described in literature (If at all?)

☐ What are the current practices?

*(Bean-planting robots in existence are made out of
5-7 subsystems:... control unit: aduinos...
Actuation: stepper motors, DC motors...)*

☐ How much is already known?

☐ Conclusion? What are the gaps?



Chapter 2: Literature review

☐ Literature review is **NOT** about selecting solutions, or describing your solution in detail. That goes into the methodology

☐ Literature review is **NOT A SUMMARY** of what appears in literature!

☐ GAPS!!!



Chapter 3: Methodology

- What? Why? How?
- Design considerations
- Design considerations: per sub-system, then **selection** of optimal solution
- Implementation of design considerations



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Chapter 3: Methodology

- ☐ How did I do the design?
Be as detailed as possible
- ✓ **Part drawings**, part circuits, evaluation tables etc. are critical in this section
- ✓ **Design calculations** should feature here. So NOT in the literature review, and NOT in the results
- ✓ **Any Finite Element Analysis** setup should be done here. The results of the calculations should be presented and discussed in Chapter Four

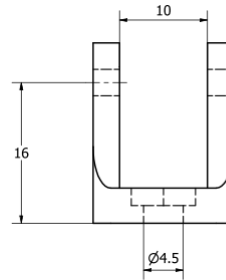
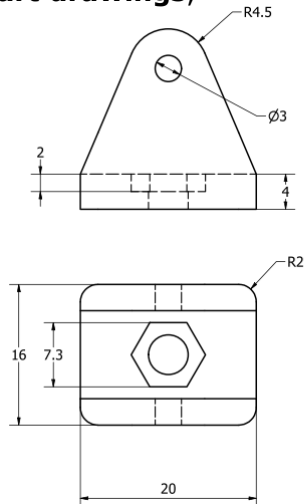


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Chapter 3: Methodology

✓ Part drawings,



No frame or Title blocks!
White background



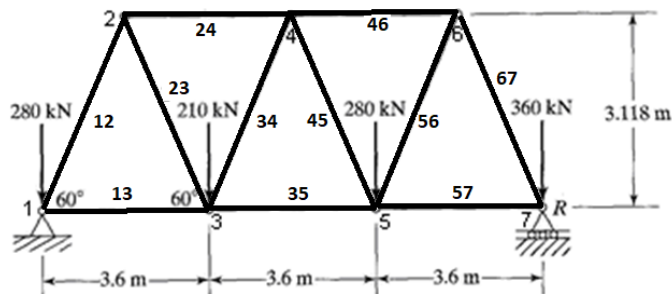
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Chapter 3: Methodology

✓ FEA Setup,

Pre-processing: Input



The seven nodes of the structure are defined as follows:

keypoint	coordinate	
	x	y
1	0	0
2	1800	3118
3	3600	0
4	5400	3118
5	7200	0
6	9000	3118
7	10800	0



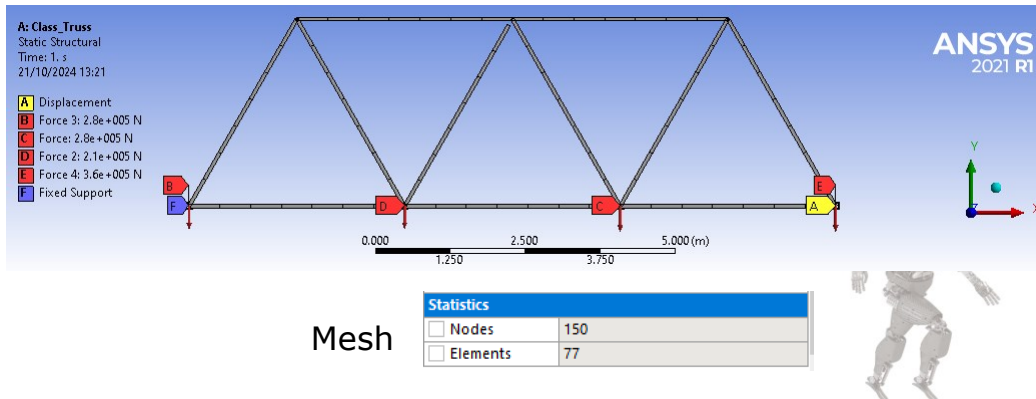
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Chapter 3: Methodology

✓ FEA Setup,

Pre-processing: Input



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Chapter 4: Results and Discussion

- ❑ What did I do (come up with)?

*Present representative data, **final detailed designs** with parts lists, budgets. **Do not repeat** drawing already in the previous chapter. Finer design details, codes can be included in the Appendix.*

- ❑ DO NOT put figures/drawings with no explanations. Results is NOT about design drawings or 3D models!
- ❑ Do NOT put calculations in the results!

Results should be enough to draw conclusions from

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Chapter 4: Results and Discussion

❑ What did I do (come up with)?

- Start with the 2D Drawings, WITH dimensions.
- Front, End and Top View
- Isometric View, labelled parts
- The 3D model should only come AFTER the 2D drawings with dimensions

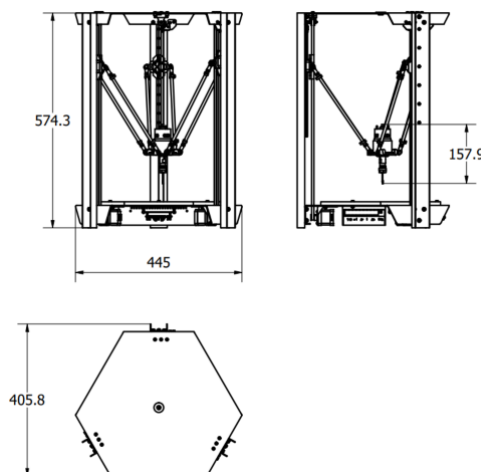


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Chapter 4: Results and Discussion

❑ What did I do (come up with)?



This is a design.

It has views and dimensions



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Chapter 4: Results and Discussion

What did I do (come up with)?

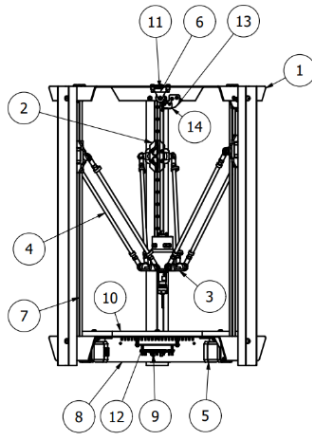


Figure 4.1: Major parts and sub-assemblies

No	QTY	Part
1	1	Mild steel frame
2	3	Precision linear guide, block and carriage assembly
3	1	Effector assembly
4	6	Arm assembly
5	3	Stepper motor assembly
6	3	Idler assembly
7	3	GT2 reinforced rubber belt
8	1	24V 10A SMPS
9	1	NUCLEO F103RB development board
10	1	Waste board
11	1	Motor cable protector
12	1	Custom prototboard CNC shield
13	3	Limit switch bracket
14	3	KW12-3 single pole double throw limit switch

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Chapter 4: Results and Discussion

What did I do (come up with)?

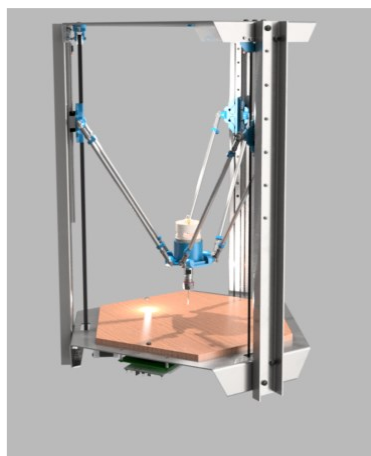


Figure 4.3: A render of the design model

This is a 3D view/model.

It should only **come AFTER** the 2D designs, with dimensions

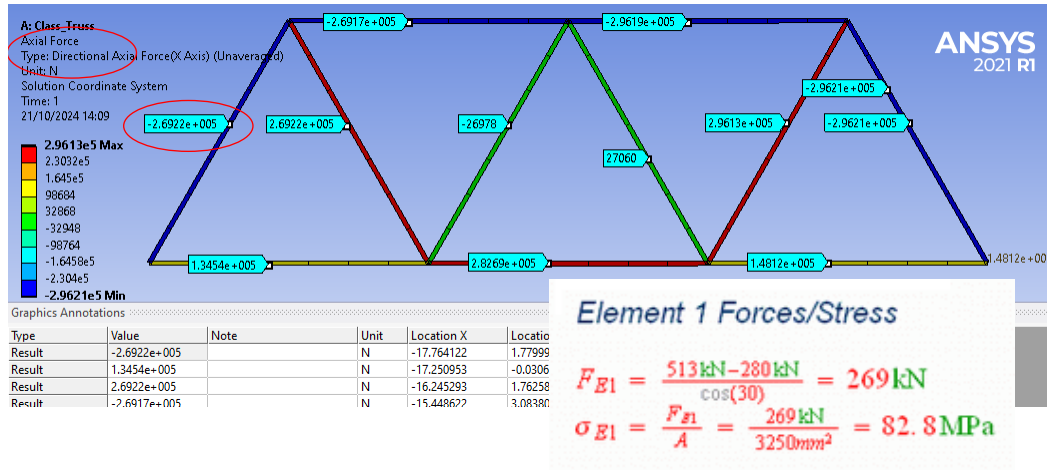


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Chapter 4: Results and Discussion

What did I do (come up with)?

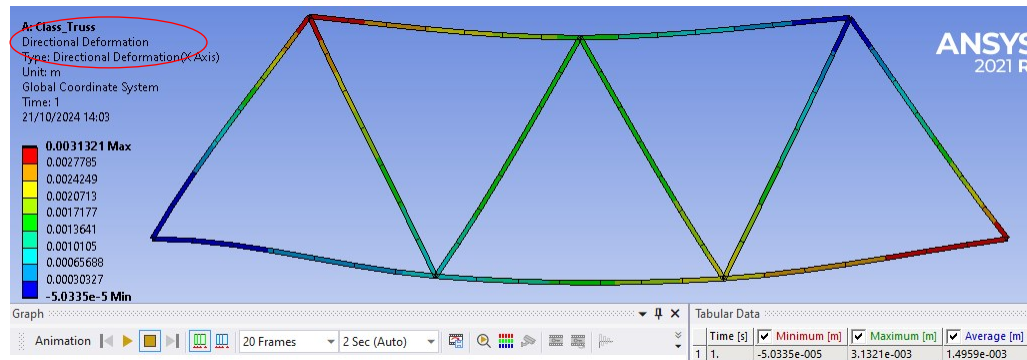


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Chapter 4: Results and Discussion

What did I do (come up with)?



Auto scale (73x)

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Chapter 4: Results and Discussion

Discussion

- ☐ What do the results mean?
- ☐ Are there any new significant results?
- ☐ Are there unexpected values?
- ☐ How does the design compare with previous/existing designs?
- ☐ Do not restate the results!



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Chapter 5: Conclusion

- ☐ What did you set out to do?
- ☐ What did you achieve?
- ☐ Was the design process completed?
- ☐ Where am I going from here? Next steps? Pending work?



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References

- ☐ Source of your information.
- ☐ Make sure that it is accurate.



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Appendices

- ☐ Time plan
- ☐ Part Drawings
- ☐ Production plans
- ☐ Programming codes
- ☐ Design reference tables, charts



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Important to Note:

The output of your first semester work (**the design**) should be such that a production technician can implement your work without much reference to you.



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Equations:

- Equations must be clear
- Numbering of equations consistent

Poor formatting

$$L=2*C+(D-d)^2/4C+\pi/2*(D+d)$$

Correct formatting

$$L = 2C + \frac{(D-d)^2}{4C} + \frac{\pi}{2}(D+d) \quad (2.7)$$

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Equations:

Correct formatting

$$\left. \begin{aligned} X &= [l_1 \cos \beta + l_2 \cos(\beta + \gamma)] \cos \alpha \\ Y &= [l_1 \cos \beta + l_2 \cos(\beta + \gamma)] \sin \alpha \\ Z &= l_1 \sin \beta + l_2 \sin(\beta + \gamma) \end{aligned} \right\} \quad (2.8)$$

Equations:

➤ An equation may be part of a sentence, e.g.,

The coordinates for the end effector of a spherical robot are obtained from the joint variables as

$$\left. \begin{aligned} X &= a \cos \alpha \cos \beta \\ Y &= a \sin \alpha \cos \beta \\ Z &= a \sin \alpha \end{aligned} \right\} \quad (2.9)$$

where the symbols carry usual meanings.

Figures:

- Must be clear
- Captions at bottom
- Numbering consistent

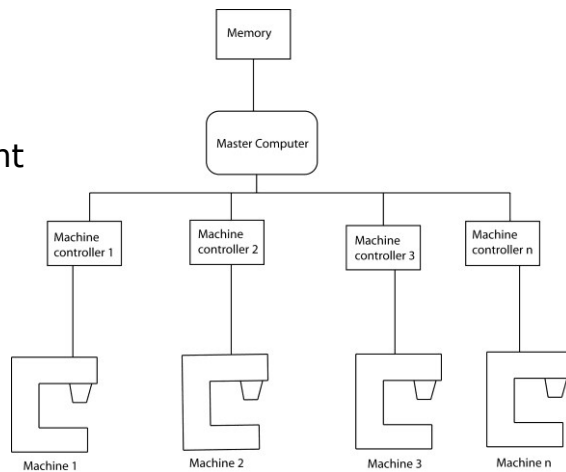


Figure 3.8 A DNC system

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Figures:

- Must be clear
- Captions at bottom
- Numbering consistent



Figure 3.9 An aircraft on final

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Figures:

- Must be clear
- Captions at bottom
- Numbering consistent

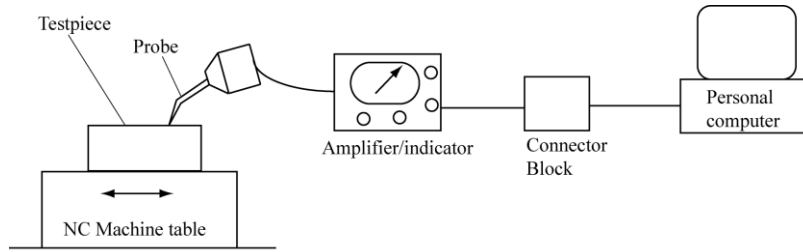


Figure 3.10 Experimental setup

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Figures:

- Must be clear
- Captions at bottom
- Numbering consistent

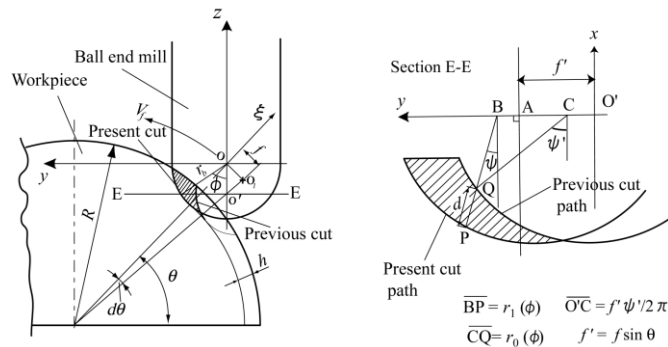


Figure. 3.11 Cutting model

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Tables:

- Must be clear
- Captions at the top
- Numbering consistent

Table. 4.2 Budget

No.	Item	Description	Qty	Cost per unit (KShs)	Total Cost (KShs)
1.	Motor	DC motor 6V, 5A	2	2,500	5,000
2.	Bearings	Linear bearings	2	3,300	6,600
3.	Fasteners	Rivets Sz18	1pkt	250	250
Total					11,750

Timeplan:

Week	Semester 1														Semester 2													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Literature Review																												
Proposal development/presentation																												
Continuous Presentations																												
Gap analysis																												
Design considerations																												
Structural Design																												
Electrical Design																												
Control Design																												
Simulation design																												
Simulations																												
Final Design																												
Interim report																												
Interim Presentation																												
Correction of interim report																												
Material Requisition																												
Structure fabrication																												
Structure assembly																												
Electrical circuit assembly																												
Control program																												
Testing of circuits and controls																												
Assembly of system																												
Testing and modification of systems																												
Final Report drafting																												
Final Presentation																												

Check points

- ☐ Why did you start?
- ☐ What did you do?
- ☐ What answer(s) did you get?
- ☐ What does it mean, anyway?



Some do's and donts

Check points:

- ☐ Originality vs Plagiarism
- ☐ Fraud
- ☐ Inconsistencies
- ☐ Blaming others
- ☐ Generalities
- ☐ etc



Finally,

Aim to finish
positively,
decisively, and
forcefully

Presentation

- ☐ Presentation done before a panel of examiners: teaching and technical staff members
- ☐ Note: It is an examination!!!
- ☐ 10 minute presentation, followed by questions from the panel
- ☐ **Time is never added.** You will be cut short if you attempt to continue with the presentation



Presentation

Required **before** the presentation:

- ☐ Two signed copies of your reports (you must have signed, and your supervisors must have signed). This is mandatory
- ☐ Your supervision cards
- ☐ A cheerful spirit
- ☐ A smile!



Presentation

- ☐ Dress appropriately for the presentation:
Official/smart casual.

Remember:

You are engineers in the making (almost ready!).

You cannot defend your work half-dressed (!), dressed in faded clothes, T-shirts with slogans, sandals/slippers, unkempt hair... etc.



Presentation

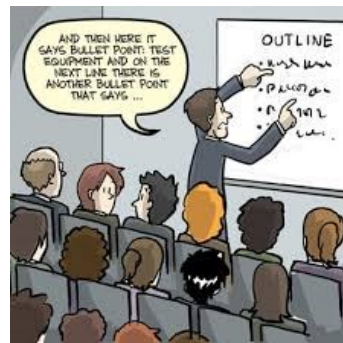
- ☐ ~~Each students presents individually~~
- ☐ ~~You may develop your unique slides or share with your project mates~~
- ☐ Stick to the template provided
- ☐ Ensure your slides are as brief as possible and to the point
- ☐ **NO PARAGRAPHS!!!** Present in point form
- ☐ **Do not cut & paste paragraphs from your report. Present points in bullet form**
- ☐ Figures, Tables and Flow charts work very well...
- ☐ Do not use any font smaller than this (Verdana 20)

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Presentation

- ☐ DO NOT READ FROM THE SLIDES
- ☐ Address your audience
- ☐ Project your voice properly: No shouting, no whispering
- ☐ Do not attempt to make your presentation on the same morning
- ☐ Go over your presentation
- ☐ Rehearse your presentation



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Interim Presentation

Time management – 15 Minutes!

1. Title slide
2. Outline
3. Intro = Background (include image of problem or existing solution) (1 slide)
4. Problem statement (1 slide)
5. Main objective and specific objectives (1 slide)
6. Literature review – a summary of literature you have looked at, tabulated. (2 slides max)
7. Gaps identified & possible approaches
8. ****Methodology – Max 6 slides – No calculations!!!**
9. Results – images, tables. Limit text and explanations (2 slides max)
10. Budget
11. Conclusion and Way forward
12. Time plan

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Presentation

****Methodology(6 slides)**

Approach = Subdivision of your project into 3-4 logical modules:

- (i) Purpose of Each module
 - (ii) Design considerations per module, possible design solutions, and design solution you are gravitating towards, and why.
 - (iii) Designs in CAD – Your Designs, not concepts!!!
- Remember: 2D, 3D Models, Isometric views, Parts
- (iv) Circuit designs

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Mark sheets:

Project Coordinator

ITEM	MAX. MARK	MARK	COMMENTS
ATTENDANCE	5		
SUPERVISION	5		
CONTINUOUS ASSESSMENT	10		
ASSESSMENT OF KEY PRESENTATIONS	10		
PROJECT PROPOSAL	15		
OTHER SUBMISSIONS	5		
TOTAL	50		

Mark sheets:

Oral Examination

ITEM	MAX. MARK	MARK	COMMENTS
PRESENTATION	Organization	3	
	Visual Aids	4	
	Clarity	3	
	Confidence	2	
	Response to questions	3	
PROBLEM DEFINITION	4		
LITERATURE REVIEW	5		
ACADEMIC SCOPE	5		
PROJECT DESIGN	10		
WORK PLAN	3		
DRAWINGS/FIGURES	3		
RESULTS & RECOMMENDATIONS	5		
TOTAL	50		

Mark sheets:

Panel Examiner

ITEM	MAX. MARK	MARK	COMMENTS
NEATNESS & LAYOUT	2		
INTRODUCTION	4		
LITERATURE REVIEW	4		
METHODOLOGY	10		
RESULTS & DISCUSSION	5		
CONCLUSION	4		
DRAWINGS/FIGURES	3		
NUMBERING (Sections, eqns., figs...)	3		
REFERENCES	3		
WORK PLAN	2		
TOTAL	40		

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Mark sheets:

Supervisor

ITEM	MAX. MARK	MARK	COMMENTS
CONTINUOUS ASSESSMENT	10		
NEATNESS AND LAYOUT	4		
LITERATURE REVIEW	6		
METHODOLOGY :	DESIGN APPROACH	5	
	CONSIDERATIONS & SELECTION	6	
	CALCULATIONS	4	
	ORIGINALITY	3	
RESULTS & RECOMMENDATIONS	5		
DRAWINGS & FIGURES	5		
NUMBERING: sections, equations, figures...	4		
WORK PLAN: MATERIALS SPECIFICATION etc.	4		
REFERENCES	4		
TOTAL	60		

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Thank you for your attention.

